GROUP PROJECT CHECKPOINT

Predicting 30-Day Customer Repeat Purchases Using Behavioural and Semantic Features

GROUP 1 Stacia Benita Student ID: 3036094289 6 November 2026

GROUP PROGRESS AND DATA AUDIT

Project question: Can a customer's merchandise transactions in the previous 90 days predict whether they will make another merchandise purchase in the next 30 days? The project uses UCI Online Retail (ID 352): 541,909 invoice lines dated 1 December 2010-9 December 2011. The raw audit found 135,080 missing customer IDs, 1,454 missing descriptions and 5,268 exact duplicate rows.

DATA CLEANING AND MY CURRENT FOCUS

- Exact duplicate invoice lines are removed, retaining one copy so the same transaction is not counted twice. Invoice and stock-code text is standardised; dates and numeric fields are parsed; descriptions are uppercased, punctuation is removed and whitespace is normalised.
- Customer IDs are not imputed. Because the task predicts repeat purchases for identifiable individuals, a row without an ID cannot be linked reliably to a customer's prior activity or future outcome. These rows are excluded from customer snapshots, model features and labels; they remain in the raw audit and are not treated as non-returning customers.
- An eligible purchase must have an identified customer, valid timestamp, positive quantity and price, and not be a cancellation. Invoice numbers beginning with "C" and negative-quantity lines are not counted as purchases. Identified merchandise returns are retained separately for return auditing, but are excluded from positive-purchase features and labels.
- Postage, carriage, manual adjustments, fees, bank charges and gift vouchers are retained for revenue auditing but excluded from product features and repeat-purchase labels because they are not merchandise purchases. Missing or unusable descriptions are excluded from product-text features.
- The notebook reports 536,641 cleaned rows; 392,692 valid identified positive-purchase lines; 391,153 merchandise lines; 1,539 transaction-charge lines; 8,506 identified merchandise-return lines; and 135,077 other invalid or unidentified lines. The last figure is a post-cleaning category - not simply the raw missing-ID count - and reflects the purchase eligibility rules.

MODELLING DESIGN AND PRELIMINARY FINDINGS

Each row represents an identified customer at a month-end snapshot. A customer is included only after at least one merchandise purchase in the preceding 90-day feature window. The label is whether they purchase merchandise in the following 30 days; target-period transactions do not enter feature calculations. There are 16,123 customer-snapshot rows across eight monthly cohorts. March-June train; July selects the model and threshold; August-October are the rolling tests. Models compare behavioural/cancellation features with TF-IDF/LSA and MiniLM product-description representations.

Evidence	Result	Interpretation
July validation selection	Boosting + LSA-TF-IDF; PR-AUC 0.615	Highest July PR-AUC; selected before final test.
August-October pooled test	PR-AUC 0.634; Lift@20% 1.77	Selected model on rolling test snapshots.
Paired test difference	+0.0009 PR-AUC; 95% CI [-0.0028, 0.0047]	Interval includes zero; no reliable text-feature gain over behavioural boosting is established.

Concrete evidence: Online_Retail_Repeat_Purchase_Project.ipynb; notebook_outputs/raw_data_audit.csv, cleaning_audit.csv, validation_results.csv, final_pooled_results.csv and paired_model_differences.csv. The notebook reports a customer-grouped bootstrap interval for pooled PR-AUC and paired model differences

INDIVIDUAL CONTRIBUTIONS

Member	Specific work completed / status	Evidence / output	Approx. hours	Next task
Stacia Benita	Assigned Stages 1-3: data audit, cleaning and invoice build. Shared outputs cover missingness, duplicate sensitivity, product/charge taxonomy and revenue audit	raw_data_audit.csv; cleaning_audit.csv; invoice_revenue_audit.csv; EDA outputs	~18 h	Independently rerun Stages 1-3; verify duplicate and charge-line decisions.

Devano Jonathan	Reviewed text-feature methodology; replaced raw sparse TF-IDF clustering with LSA-TF-IDF; checked matched K=16 and MiniLM clusters; reviewed October errors; prepared checkpoint.	cluster_quality.csv; cluster_matched_k_quality.csv; product_cluster_examples.csv; error_analysis_summary.csv; 19 tests	~20 h	Accept cluster descriptions and review final wording.
Surjadjajadi Clara Natalie	Completed shared snapshot, model and uncertainty outputs; final pipeline review and report integration remain.	customer_snapshot_features.csv; validation_results.csv; rolling_monthly_results.csv; paired_model_differences.csv	~19 h	Verify pipeline section and merge final report.

NEXT STEPS

Target	Owner	Task
6-7 Nov 2026	All members	Finish personal evidence checks; submit each individual two-page checkpoint with confirmed hours
9-14 Nov 2026	All members	Reconcile sensitivity and cluster-count results, finalise figures and maintain iteration log.
15-20 Nov 2026	All members	Merge report sections with consistent claims and caveats.
21-25 Nov 2026	All members	Build slides, rehearse and check numerical consistency.

BLOCKERS / SUPPORT NEEDED

Each member must confirm their own approximate hours and personally verify assigned evidence. Cluster labels still need human acceptance.

AI USE AND VERIFICATION

Some group members used Codex and others used GPT-Luna 6 to support the project. Codex was also used to assist with data analysis and to generate Python code for the analyses and visualisations. AI assistance helped review the data-cleaning process, check for potentially missed issues or relevant records, and refine the reasoning for including or excluding data.

AI-generated suggestions and code were reviewed against the archived dataset and saved result tables. The notebook was run end to end, and the project's 19 contract tests passed. Group members remain responsible for verifying their assigned analyses, interpreting results and confirming that the final claims are supported by the evidence.